Worksheet 2-8 & 2-9 A Product and Quotient Rules #1
AP Calculus Name_____

Directions: Find the derivatives of the given functions below. Use any method.

1. $f(x) = 2x^3 \cos(x)$

2. $g(x) = e^x\sqrt{x}$

3. $y = x^2 \ln(x) - x^3$

4. $h(x) = 4x - x^2e^x + 5\sin(x)$

5. $y = (3x - 1)(4x^2 + 5)$

6. $k(x) = -\frac{1}{x} - 3\sin(x) - x^4\cos(x)$

7. $h(x) = \frac{4x - 1}{3x + 2}$

8. $g(x) = \frac{\sin x}{x}$

9. $f(x) = \dfrac{x^3 + 2x^2 - x}{2x}$

10. $m(x) = \dfrac{4x}{\ln x}$

11. $y = \dfrac{3x^4 - 2x^3 - 3\sqrt{x}}{x}$

12. $y = \dfrac{2x^5}{\cos x}$

13. $f(x) = \dfrac{e^x}{4\sin x}$

14. $g(x) = \dfrac{2x + 4}{3x + 2}$

15. $y = \dfrac{x^3 + 3x^2 - x}{x^2}$